

A project report on

SEPSIS DETECTION AND CLASSIFICATION USING IMPROVED RANDOM FOREST

Submitted in partial fulfillment for the award of the degree of

M.Tech [Integrated]

**Computer Science and Engineering with
Specialization in Data Science**

by

MADHURIKA M (20MID0208)

Under the Guidance of

Dr. RAMESH C

Associate Professor Grade 1



VIT[®]
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I here by declare that the thesis entitled “**SEPSIS DETECTION AND CLASSIFICATION USING IMPROVED RANDOM FOREST**” submitted by me, for the award of the degree of M.Tech [Integrated] Computer Science and Engineering with Specialization in Data Science is a record of bonafide work carried out by me under the supervision of Dr. Ramesh C.

I further declare that the work reported in this thesis has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Place: Vellore

Date: 16.04.2025

Signature of the Candidate

CERTIFICATE

This is to certify that the thesis entitled “ SEPSIS DETECTION AND CLASSIFICATION USING IMPROVED RANDOM FOREST” submitted by MADHURIKA M (20MID0208), School of Computer Science and Engineering, Vellore Institute of Technology, Vellore for the award of the degree M.Tech [Integrated] Computer Science and Engineering with Specialization in Data Science is a record of bonafide work carried out by her under my supervision.

The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university. The Project report fulfils the requirements and regulations of VELLORE INSTITUTE OF TECHNOLOGY, VELLORE and in my opinion meets the necessary standards for submission.

Signature of the Guide

Signature of the HoD

Internal Examiner

External Examiner

ABSTRACT

Due to the high death rates and serious consequences that can result from delayed treatment, early diagnosis of sepsis has gained significant attention in medical research in recent years. Promising strategies for detecting sepsis early on are provided by machine learning and sophisticated statistical techniques. The goal of this proposed work is to present a thorough analysis of sepsis detection algorithms, highlighting both conventional techniques and more recent developments. In addition to detection, this proposed work addresses the common problem of class imbalance in sepsis datasets, enhances model performance through feature engineering and optimisation techniques, and examines important problems related to handling missing data in electronic health records (EHRs). A number of tactics are covered to improve prediction accuracy and clinical applicability, such as class resampling techniques, ensemble learning, and imputation methods. A vital resource for researchers and doctors alike, the project also identifies future prospects in sepsis detection research, including real time prediction and the incorporation of explainability tools in model building.

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CONTENTS

Title	Page No
List of Tables	xi
List of Figures	x
1. Introduction	1
1.1. Theoretical Background	1
1.2. Motivation	1
1.3. Aim of the proposed Work	2
1.4. Objective of the proposed work	2
2. Literature Survey	3
2.1. Survey of the Existing Models	3
2.2. Gaps Identified in the Survey	8
2.3. Problem Statement	9
3. Overview of the Proposed System	10
4. Requirements Analysis and Design	11
4.1. Requirements Analysis	11
4.1.1. Functional Requirements	11
4.1.1.1. Product Perspective	11
4.1.1.2. Product Features	11
4.1.1.3. User Characteristics	12
4.1.1.4. Assumption & Dependencies	12
4.1.1.5. Domain Requirements	12
4.1.2. Non-Functional Requirements	13
4.1.3. System Modeling	13
4.1.4. Engineering Standard Requirements	14
4.1.5. System Requirements	14
4.1.5.1. Hardware Requirements	14
4.1.5.2. Software Requirements	15

4.2. System Design	15
4.2.1. System Architecture	15
4.2.2. Detailed Design	17
5. Implementation and Testing	18
6. Results and Discussion ..	24
7. Conclusion and Scope for Future Work.....	31
Annexure – I - Pseudo Code	32
References	35

LIST OF TABLES

Title	Page No.
Table 1.1 – Litration survey of existing works	3

LIST OF FIGURES

Title	Page No.
Fig. 1 Image of ER Diagram of proposed sepsis prediction model	13
Fig. 2 Image of Data Flow diagram of proposed prediction sepsis model	17
Fig. 3.1 Code Implementation – dataset loading	21
Fig. 3.2 Code Implementation – data preprocessing	22
Fig. 3.3 Code Implementation – model training and saving	23
Fig. 4 Before and after imputation of missing values	26
Fig. 5 Result of handling class imbalance	27
Fig. 6 Result of the evaluation metrics	27
Fig. 7 Feature importance in bar chart	27
Fig. 8 AUC-ROC Curve	28
Fig. 9 Implementation of the webpage	29
Fig. 10 Test values in webpage	29
Fig. 11 Submitting test values in webpage	30
Fig. 12 Final prediction results	30

1. INTRODUCTION

1.1 BACKGROUND

Sepsis is a serious illness that causes tissue damage, organ failure, and often death. It is brought on by the body's overreaction to an infection. About 48.9 million cases and 11 million deaths globally are attributed to sepsis each year, making up over 20% of all fatalities. A significant health burden still exists, especially in low- and middle-income nations, where 85% of sepsis cases take place. In many areas, it is linked to limited access to healthcare, problems for newborns, and infections in mothers.

Sepsis's quick progression presents a significant treatment problem. According to studies, the risk of dying rises by 4-9% for every hour that treatment is put off in the early stages. Even though sepsis is a serious illness, it can be challenging to diagnose early because of its vague symptoms and the fact that each patient experiences it differently. This frequently results in delayed therapy, which has a negative impact on survival chances. Conventional diagnosis techniques mostly rely on test results and clinical assessments, which can be laborious and unreliable, particularly in environments with limited resources.

Nonetheless, the growing accessibility of data from electronic health records (EHRs) has made it possible to use machine learning methods to enhance sepsis early detection. Despite these developments, it has been difficult to create efficient models for sepsis prediction due to a number of obstacles, including missing data, class imbalance, and high dimensionality of features. In order to improve sepsis management worldwide and lower the startling fatality rates, these issues must be resolved.

1.2 MOTIVATION

The urgent need to stop the global sepsis pandemic is what inspired this research. Although sepsis can strike anyone at any age, it is most fatal for those who are most at risk, including elderly individuals, pregnant women, and newborns. An estimated 1.2 million children and 3 million infants worldwide are predicted to die from sepsis each year, with a significant number of deaths taking place in low-income nations as a result of delayed diagnosis and treatment. Furthermore, 61,000 people die from maternal sepsis each year, mostly in areas with poor access to healthcare.

Machine learning has enormous potential to help with sepsis prediction and early detection given the abundance of clinical data already available. However, problems including insufficient data and imbalanced datasets, which are widespread in global health settings, make it difficult to create appropriate models. Insufficient data and sepsis patients are greatly outnumbered by non-sepsis cases. Methods that address class imbalance with ADASYN, handle missing data with sophisticated imputation techniques, and combine Random Forest with LightGBM provide a path ahead. Through enhancing the accuracy and dependability of sepsis prediction models, this project seeks to give medical professionals worldwide a vital tool to identify sepsis sooner, save lives, and lower healthcare expenses.

1.3 AIM OF THIS PROJECT

This project's scope includes creating and deploying cutting-edge machine learning models for the early diagnosis of sepsis, with an emphasis on improving predictive accuracy and usefulness in clinical contexts. Obtaining and preparing clinical data from various healthcare sources, feature engineering to extract important sepsis indicators, and investigating various machine learning techniques to develop reliable predictive models are important fields. Class imbalance, missing values in the dataset, and the use of hyperparameter adjustment for validation are further issues that the project tackles.

1.4 OBJECTIVES

This project's main goal is to create a machine learning model for the early diagnosis of sepsis through the analysis of clinical data and electronic health records (EHR). It seeks to improve forecast accuracy by tackling issues like class imbalance and missing data using sophisticated methods including SMOTE, feature engineering, and hybrid model creation (Random Forest and LightGBM). In order to improve patient outcomes and early sepsis detection, the project also aims to optimise model performance through hyperparameter tuning and develop a useful, real world tool that healthcare providers can utilise.

2. LITRATURE REVIEW

2.1 SURVEY OF THE EXISTING WORKS

S NO	TITLE	YEAR	SUMMARY	LIMITATIONS
1	HRV -Based Regression Analysis in Newborns With Sepsis: Forecasting BAYLEY-III Scores at 6 and 12 Months	2024	The study employs Artificial Intelligence (AI) regression models to predict neurodevelopmental outcomes based on Heart Rate Variability (HRV) measurements from preterm babies with and without sepsis. These models were verified using a Leave One Subject Out (LOSO) framework, and the HRV data, which was obtained from ECG signals recorded at corrected term age, was used as input features. The models were able to predict BAYLEY-III cognition and language scores at 6 and 12 months with a Mean Absolute Error (MAE) of fewer than 5 points. The results point to a connection between the maturation of the central nervous system and the development of the autonomic nervous system, suggesting that HRV features in conjunction with AI models can accurately predict neurodevelopmental changes in preterm infants.	Limited Sample Size Single-Center Data Potential Biases in HRV Measurements Generalizability of Findings Influence of Confounding Factors Model Interpretability
2	Advancing Early Detection of Sepsis With Temporal Convolutional Networks Using ECG Signals	2024	Sepsis is a potentially lethal illness with high death rates, particularly in intensive care units, because of delayed diagnosis and treatment. Traditional clinical scoring techniques like SIRS, qSOFA, and MEWS are inadequate for predicting early sepsis. This work proposes a machine learning classifier that can detect sepsis up to six hours before a clinical diagnosis is made using patient demographics, vital signs, and electronic medical data. A novel mean median data imputation technique based on outliers is used to increase prediction accuracy. Of the four algorithms that were tested Random Forest, Logistic Regression, Gradient Boosting, and Decision Tree Random Forest yielded the best results, with an accuracy of 99.01%.	Limited Quantity of Sepsis Cases Overfitting Tendency Early Convergence Behavior Restricted Analysis of Control Group Recordings Need for External Validation Challenges of Multicenter Studies
3	Sepsis Prediction Using Machine Learning	2023	Sepsis, a dangerous illness with a high death rate, especially in patients in intensive care units, is characterised by a broad inflammatory response to infection. Early detection and treatment are crucial. This study explores the use of machine learning models, specifically the Random Forest	Variability in Sepsis Definitions Data Heterogeneity Class Imbalance

			Classifier, to improve sepsis detection using laboratory results, vital signs, and electronic medical data. By comparing several classifiers, the work aims to increase detection accuracy and efficiency while addressing the drawbacks of existing models like LSTMs and RNNs. The improved accuracy and potential for early and accurate sepsis diagnosis of the proposed technique may lead to better patient outcomes.	Model Interpretability External Validation Real Time Integration
4	Machine Learning-Based Early Prediction of Sepsis Using Electronic Health Records: A Systematic Review	2023	The review, which examined 42 papers from databases such as PubMed, IEEE Xplore, Google Scholar, and Scopus, emphasizes that the majority of the studies are retrospective and mostly concentrate on data from the US. With different quality and performance criteria like AUROC, sensitivity, and specificity, the models make use of a variety of datasets and sepsis definitions. Important results indicate that ML/DL techniques can forecast the start of sepsis 26 hours ahead of time, which could improve patient outcomes and early intervention. But there are still issues with model interpretability, class imbalance, and data heterogeneity. Enhancing data quality, external validation, integrating real-time prediction, and establishing clinical benefit are some future goals.	Variability in Sepsis Definitions Data Heterogeneity Class Imbalance Model Interpretability External Validation Real Time Integration
5	Comparing artificial intelligence strategies for early sepsis detection in the ICU: an experimental study	2023	The study focuses on how various data imputation techniques affect missing values that are frequently present in clinical datasets. The study assesses five machine learning techniques and five deep learning models through a two-phase experiment, calculating the area under the precision-recall curve (AUPRC) to gauge each method's performance. The study offers open source code to improve repeatability and comparability in an effort to create a standard foundation for further sepsis prediction research.	Data Imputation Generalizability Computational Complexity Interpretability Dataset Bias Early Prediction Challenge Real Time Implementation
6	A Comprehensive Machine Learning Based Pipeline for	2022	Sepsis is a potentially lethal illness with high death rates, particularly in intensive care units, because of delayed diagnosis and treatment. Traditional clinical scoring techniques like SIRS, qSOFA, and MEWS are inadequate for predicting early sepsis. This work proposes a machine learning classifier	Operational Constraints in Clinical Settings Variability in Data Collection

	an Accurate Early Prediction of Sepsis in ICU		that can detect sepsis up to six hours before a clinical diagnosis is made using patient demographics, vital signs, and electronic medical data. Of the four algorithms that were tested Random Forest, Logistic Regression, Gradient Boosting, and Decision Tree Random Forest yielded the best results, with an accuracy of 99.01%. The study highlights the significance of early sepsis detection for improving patient outcomes and reducing hospital costs.	Dependence on Data Quality Challenges with Data Imputation Techniques Limitations of Logistic Regression and Gradient Boosting Challenges in Handling Imbalanced Data
7	A Deep Learning-Based Sepsis Estimation Scheme	2021	The goal of the project is to use electronic health records (EHRs) to create a machine learning (ML) method for predicting severe sepsis and septic shock. High accuracy (93.84%), sensitivity (93.22%), and specificity (95.25%) were attained by the machine learning model after it was trained and validated on patient data containing certain ICD codes and blood culture results. While there were no appreciable variations in alert values amongst ICU wards, the application of this ML model resulted in a minor rise in IV usage and lab testing. Traditional rule-based systems are insufficient for detecting sepsis, a serious medical illness that needs early detection for better results. By managing a variety of data kinds and recording times, the suggested deep learning framework which makes use of CNN and LSTM networks offers superior prediction capabilities.	Data Limitations Generalizability Real Time Implementation Data Privacy and Security Computational Resources
8	Early detection of Sepsis in critical patients using Random Forest Classifier	2020	The complexity of the illness and the problem of missing laboratory data make it difficult to predict sepsis in intensive care unit patients. The authors employed constant imputation with zeros and the last observation carried forward (LOCF) approach to deal with missing data. They used cost-sensitive learning to address class imbalance and used ANOVA testing to choose 19 out of 40 features. The data was used to train a random forest classifier, which obtained utility scores of around 57% and 22% on two datasets from the 2019 PhysioNet/Computing in Cardiology (CinC) competition. Despite being superior to previous methods, the model still faced problems with data quality, computational cost, and generalisability. The study highlights the	Data Availability and Quality Computational Complexity Generalizability Interpretability

			need for additional research to increase the predicted accuracy and resilience of the model.	
9	Practical Machine Learning-Based Sepsis Prediction	2020	Because of its vague symptoms and lack of conclusive diagnostics, sepsis is a serious, costly, and difficult condition to identify. Improving patient outcomes requires early detection. This work uses a variety of feature sets, such as vital signs, lab data, and expert inputs, to assess how well machine learning models more especially, XGBoost predict sepsis. The results show that models that include both laboratory and extensive clinical data outperform models that only use vital signs. Optimizing prediction algorithms using readily available data is still crucial, though, because a large number of sepsis cases start outside of hospitals. Personalized and useful machine learning techniques that are adapted to certain populations and the data at hand may improve the management and prediction of sepsis.	Dependency on Clinical Expert and Laboratory Biomarker Features Effectiveness of Vital Sign Measures Alone Variation in Performance Across Different Patient Populations Constraints of Data Availability in Non-ICU Settings False Positive Rates and Precision in Sepsis Prediction
10	A new effective machine learning framework for sepsis diagnosis	2018	This study focusses on the issue of early detection and effective assessment of sepsis, a potentially lethal condition. By combining metabolic data from sepsis patients with gas chromatography mass spectrometry (GC-MS), the researchers developed a machine-learning framework employing kernel extreme learning machine (KELM) optimised using chaotic fruit fly optimisation algorithm (CFOA). This approach had an accuracy of 81.6% in diagnosing sepsis, with 89.57% sensitivity and 65.77% specificity. The study also discovered five significant biomarkers associated with sepsis, including lower levels of linoleic acid and cholesterol and increased acetic acid. The proposed RF-CFOA-KELM model outperformed other methods and offers a potentially helpful tool for clinical decision support in the diagnosis of sepsis.	Limited Sample Size Data Preprocessing Challenges Generalizability to Other Populations Lack of External Validation
11	Improving Prediction Performance Using Hierarchical Analysis of Real-Time Data: A	2018	This work introduces a novel hierarchical machine learning method that enhances patient early sepsis diagnosis by utilising high frequency physiological data. The method makes use of a multi-layer classifier system, where the first layer predicts sepsis based on continuous physiological data, and the second layer refines predictions based on the probabilistic	Data Collection and Quality Generalizability and External Validation Computational Complexity

	Sepsis Case Study		output of the first layer. The algorithm demonstrated significant improvements in predicting sepsis sooner than traditional methods like Systemic Inflammatory Response Syndrome (SIRS) criteria, with missed predictions dropping from 11.76% to 3.21% and sepsis being predicted on average 204.87 minutes earlier. By combining real time data analysis with hierarchical modelling, the approach effectively improves prediction accuracy and enables timely interventions in critical care scenarios.	Interpretability of Predictions Integration with Clinical Workflow
12	Early Sepsis Detection in Critical Care Patients Using Multiscale Blood Pressure and Heart rate Dynamics	2017	The application of high resolution heart rate (HR) and blood pressure (BP) dynamics for the early identification of sepsis in intensive care unit (ICU) patients is examined in this work. Sepsis is one of the most common causes of death, and postponing treatment significantly raises the risk of death. Traditional methods rely on clinical data from electronic medical records (EMR), but often lack the clarity needed for early prediction. This study forecasts sepsis four hours in advance using high frequency physiological data in an attempt to reduce sepsis-related morbidity and mortality. Researchers demonstrated that these dynamic markers can independently predict the onset of sepsis earlier than conventional methods by integrating patient demographics, clinical history, blood pressure, and heart rate data into a multivariate model.	Sample Size and Cohort Specificity Predictive Model Variability and Generalizability
13	SIRS, qSOFA and new sepsis definition	2017	In comparison to the Systemic Inflammatory Response Syndrome (SIRS) criteria, the study assesses how well the Sepsis-3 definitions in particular, the fast SOFA (qSOFA) score diagnose sepsis and forecast patient outcomes. They discovered that although qSOFA has a high specificity, it has a low sensitivity for identifying early sepsis, which could cause it to overlook important early-stage patients. The study also demonstrated that while SIRS and qSOFA are equally accurate at predicting organ dysfunction, SIRS occasionally exhibits superior predictive accuracy for in-hospital mortality. More research is required to improve sepsis identification and management techniques, according to the study, which also highlights the practical difficulties of applying Sepsis-3 definitions in clinical practice,	Sensitivity of qSOFA Score Early Detection of Sepsis Implementation Challenges Accuracy of SIRS vs. qSOFA in Predicting Mortality

			such as the requirement for protocol revisions and provider retraining.	
14	Remote Patient Monitoring for Improving Outpatient Care of Patients at Risk for Sepsis	2016	In order to reduce hospital readmissions and mortality rates and enhance the standard of outpatient care, the project aims to develop a remote monitoring platform that monitors sepsis markers. Sepsis, a major contributor to hospital readmissions and exorbitant medical costs, is often the consequence of inadequate postoperative care. While several indications, such infection, heart rate, respiration rate, blood pressure, and fever, may be monitored at home, there are significant challenges, like accounting for the confounding effects of physical activity and the unpredictability of biometric sensors.	Patient Activity Confounding Data Baselining Vitals Patient Involvement and Compliance Quality of Biometric Devices Communication of Findings
15	The Prevention and Handling of the Missing Data	2013	By dividing missing data into three categories Missing Completely at Random (MCAR), Missing at Random (MAR), and Missing Not at Random (MNAR) this paper provides a comprehensive review of missing data management strategies. Kang emphasises how important it is to properly handle missing data in order to avoid weakened statistical power, skewed parameter estimations, and degraded sample representativeness. Listwise deletion, pairwise deletion, mean substitution, regression imputation, and multiple imputation are the techniques used to deal with missing data.	-

(Table 1.1 – Literature survey of existing works)

2.2 GAPS IDENTIFIED

- **Incorporation of Additional Data Sources:** Including additional relevant data from clinical sources, including genetic information, patient history, or electronic health records (EHR), could increase the model's capacity for prediction. This could make it easier to get more detailed information on the patient's health.
- **Advanced Imputation Techniques:** Exploring more sophisticated imputation techniques beyond Last Observation Carried Forward (LOCF) and constant imputation, such as multiple imputation or deep learning based imputation methods, could improve handling of missing data and potentially lead to better model performance.

- **Feature Engineering:** Further research into feature engineering could help identify and create new features that are more predictive of sepsis. This could involve creating time series features, aggregating trends, or utilizing domain-specific knowledge to derive more informative variables.
- **Exploring Different Machine Learning Models:** While Random Forest is effective, experimenting with other machine learning models, like gradient boosting machines, deep neural networks, or ensemble methods, could yield better results, especially in handling class imbalance and complex interactions between features.
- **Handling Class Imbalance with Advanced Techniques:** In order to improve the model's performance on minority classes, future research should concentrate on using more sophisticated methods for addressing class imbalance, such as SMOTE (Synthetic Minority Over-sampling Technique), ADASYN (Adaptive Synthetic Sampling), or further investigating cost-sensitive learning.

2.3 PROBLEM STATEMENT

Current sepsis prediction models face significant challenges in clinical application due to class imbalance, where rare but critical sepsis cases are often misclassified, leading to compromised patient outcomes. Additionally, traditional methods for dealing with missing data, including Last Observation Carried Forward (LOCF), are not very reliable and can cause bias. Additionally, the complex and dynamic character of sepsis progression is not adequately captured by current feature engineering techniques. The development of precise and trustworthy predictive models for early sepsis identification is hampered by these constraints taken together.

3. OVERVIEW OF THE PROPOSED SYSTEM

In order to facilitate real time clinical decision making, the suggested approach focusses on creating a reliable, ensemble based classification model to perform early sepsis detection and classification that is incorporated into an intuitive web application. The model integrates the advantages of LightGBM (LGBM) and Random Forest (RF) in a stacking ensemble framework to improve accuracy, manage unbalanced data, and guarantee computational efficiency.

The pipeline begins with data preprocessing, where irrelevant columns are removed, and missing values are addressed using advanced techniques Iterative Imputer to numerical features and mode imputation to categorical features. Feature names are standardized to ensure compatibility with the modeling framework, and the target variable SepsisLabel is defined for classification. ADASYN (Adaptive Synthetic Sampling) is used to create synthetic samples for the minority (sepsis) class in order to address the serious problem of class imbalance that frequently occurs in medical datasets. As a result, the model is better able to identify uncommon but significant cases and is less biased towards non-sepsis data.

An 80:20 stratified ratio is then used to divide the dataset into training and validation sets. Random Forest is used as the base model in a stacking classifier because of its capacity to handle complicated, high dimensional data, while LightGBM is used as the meta model because of its efficiency, scalability, and speed when dealing with imbalanced data. To prevent overfitting, the meta model is only trained using the original model's predictions. Accuracy, precision, recall, F1-score, ROC-AUC, and feature importance for the columns by the model are among the performance metrics used in the model evaluation process. This ensures a comprehensive evaluation of the classifier's dependability and efficacy in detecting sepsis cases.

To make the solution accessible and practical for clinical use, the trained model is deployed as part of a web application. This application provides a clean, interactive, and intuitive interface where users (e.g., medical professionals) can enter input patient data and view real time predictions on sepsis risk. The target is to enhance clinical decision making by offering a fast and accurate sepsis risk evaluation tool powered by machine learning.

4. REQUIREMENTS ANALYSIS AND DESIGN

4.1 REQUIREMENTS ANALYSIS

4.1.1 FUNCTIONAL REQUIREMENTS

The system must process patient medical data and use a trained machine learning model for predict sepsis risk. It should display real time risk levels and alerts through a web based clinical dashboard for healthcare professionals.

4.1.1.1 PRODUCT PERSPECTIVE

The Sepsis Early Detection System is a Flask based web application that uses machine learning for predict sepsis risk in ICU patients. It processes patient medical data collected from hospital monitoring systems and laboratory tests, providing real time risk assessments through an intuitive clinical dashboard. The system analyzes vital signs, blood test results, and patient history to generate early warnings for potential sepsis cases.

4.1.1.2 PRODUCT FEATURES

Data Preprocessing: Handles missing data using Iterative Imputation and removes irrelevant features.

Class Imbalance Handling: Uses ADASYN (Adaptive Synthetic Sampling) to balance sepsis and non-sepsis cases.

Real Time Prediction: Deploys a trained model to predict sepsis risk based on incoming patient data.

Web Application: An easy to use web application that uses FLASK framework in python to classify and predicte sepsis using the trained model.

4.1.1.3 USER CHARACTERISTICS

Clinicians & ICU Staff: Primary users who rely on the system for early sepsis alerts.

Data Scientists/Researchers: Secondary users who may fine tune the model or analyze trends.

Hospital Administrators: Stakeholders interested in improving patient outcomes and reducing sepsis related mortality.

4.1.1.4 ASSUMPTIONS AND DEPENDENCIES

Assumptions:

- Data is available in a structured format.
- Missing data follows a Missing at Random (MAR) pattern.
- The model generalizes well across different ICU settings.

Dependencies:

- Availability of Python based ML libraries (Scikit-learn,Pandas,Numpy,Simple Imputer, LightGBM, Imbalanced-learn).

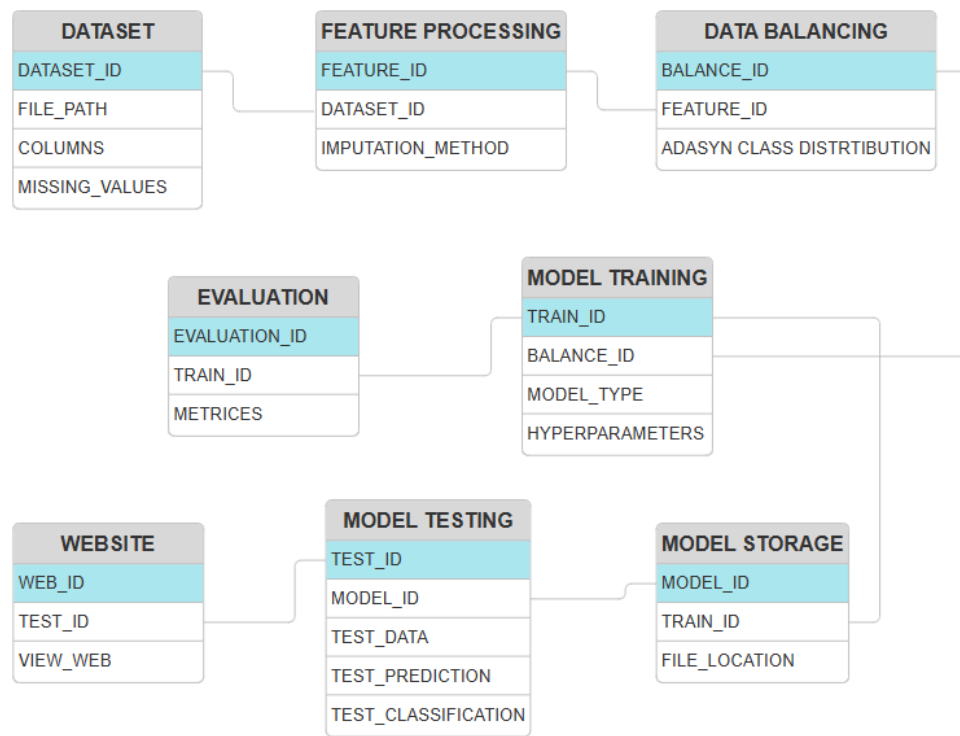
4.1.1.5 DOMAIN REQUIREMENTS

- Data Privacy and Security: The system must ensure the confidentiality and protection of sensitive patient data throughout its lifecycle.
- Timely Response: The model should generate predictions quickly enough to support effective clinical decision-making in real world scenarios.
- Model Transparency: The output and reasoning of the model should be understandable and interpretable to medical professionals to support trust and informed usage.

4.1.2 NON FUNCTIONAL REQUIREMENTS

- Interoperability – The system should integrate with existing hospital databases and EHR systems.
- Auditability – All model decisions and user interactions should be logged for future audits.
- Portability – The system should run on different platforms (cloud, on-premises, or edge devices).
- Energy Efficiency – The model should optimize resource usage to minimize computational costs.
- Extensibility – The architecture should allow adding new features or ML models easily.

4.1.3 SYSTEM MODELING



(Fig. 1 : Image of ER Diagram of proposed sepsis prediction model)

The above image is an ER (Entity-Relationship) diagram of the proposed machine learning model for sepsis detection and classification and its deployment via a user friendly web application. Each entity is connected through foreign key relationships, showcasing the flow of

data from raw input to model prediction and web based output. The design ensures modularity, efficient tracking of processes, and scalability for real world healthcare deployment.

4.1.4 ENGINEERING STANDARD REQUIREMENTS

- Follow well defined software development lifecycle practices to ensure clarity, traceability, and maintainability.
- Ensure thorough validation and verification of the system to maintain accuracy and reliability in medical decision making.
- Apply robust testing procedures, including unit testing, integration testing, and system-level testing, to verify the performance of the ML model in clinical scenarios.
- Implement secure and ethical coding practices to protect sensitive medical data and ensure compliance with data privacy principles.
- Maintain proper documentation throughout the development process for traceability, auditability, and ease of future updates or improvements.

4.1.5 SYSTEM REQUIREMENTS

4.1.5.1 HARDWARE REQUIREMENTS

- Processor - Dual Core Processor 2.6.0 GHZ
- RAM - 8 GB
- Hard Disk - 160 GB
- Compact Disk - 650 Mb
- Keyboard - Standard Keyboard
- Monitor - 15 Inch Color Monitor

4.1.5.2 SOFTWARE REQUIREMENTS

- Operating system - Windows 10
- Front End - HTML, CSS (rendered using FLASK)
- Back end - PYTHON
- Tool - Python 3.7
- IDE - Android Studio

4.2 SYSTEM DESIGN

4.2.1 SYSTEM ARCHITECTURE

The suggested sepsis detection model's system architecture is divided into multiple interrelated levels that cooperate to provide precise prediction and real time usability through a web application.

The first layer, Data Collection and Sources, involves gathering data, which includes vital signs, lab results, demographics, and medical history. This unprocessed dataset serves as the basis for further modelling and processing.

In the Data Preprocessing Layer several procedures guarantee the accuracy and applicability of the data. Data Cleaning is performed using advanced imputation techniques such as Multiple Imputation (MI) to handle missing values, thus preserving the dataset's integrity and statistical power. Outlier detection and removal are also conducted to enhance data quality. Following this, Feature Engineering is applied to extract meaningful insights. Domain-specific features are created using medical expertise such as combining vital signs with demographic details while temporal features are generated using lagged variables to capture trends over time. This helps in the early identification of sepsis progression.

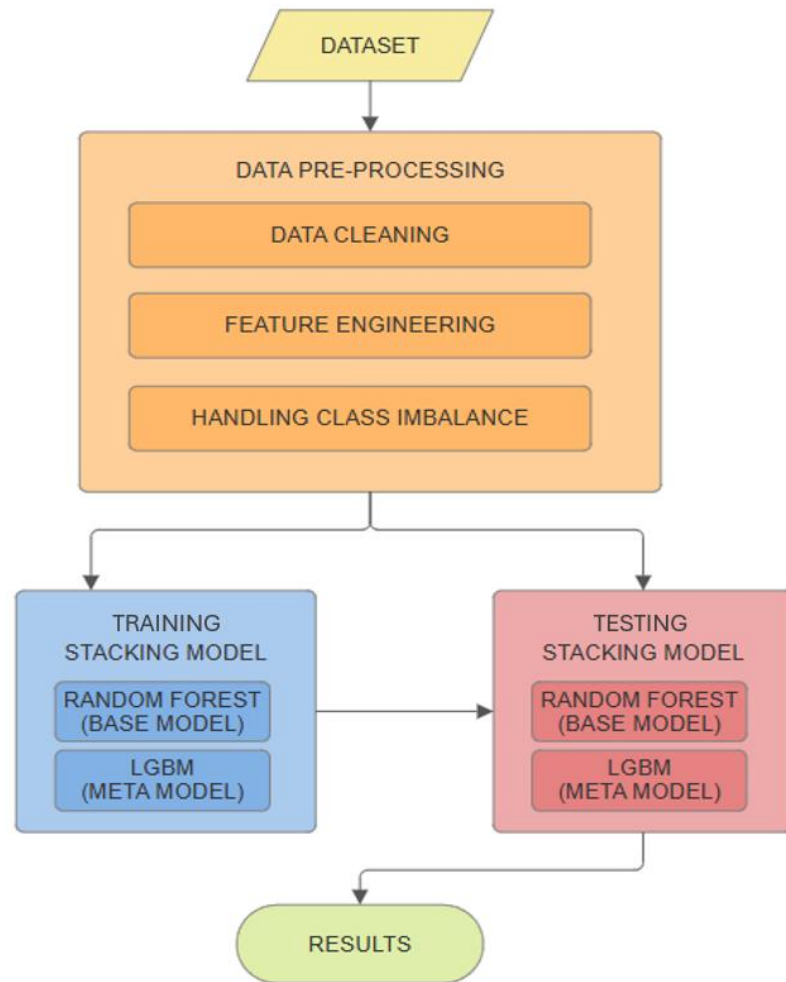
The Modeling Layer employs an ensemble architecture that integrates Random Forest and LightGBM. Random Forest is employed due to its capacity to handle complicated feature correlations and non-linear interactions, whereas LightGBM is chosen due to its speed, effectiveness, and capacity to manage big and unbalanced datasets. When these two models are

combined, stability and prediction accuracy are improved. Synthetic oversampling approaches like as SMOTE (Synthetic Minority Over-sampling Technique) and ADASYN (Adaptive Synthetic Sampling) are used to further address the problem of class imbalance, which is prevalent in sepsis datasets. By creating artificial instances of the minority class, these techniques greatly enhance the model's capacity to identify uncommon cases of sepsis.

The Model Evaluation and Interpretation Layer is crucial for performance assessment and clinical relevance. Several metrics, such as accuracy, precision, recall, F1-score, AUC-ROC, and log loss, are used to assess the model. To guarantee prompt sepsis detection, special focus is paid to lowering false negatives. Furthermore, feature importance scores are provided by Random Forest and LightGBM, and they are examined to determine the impact of specific factors. This helps clinicians identify important risk factors linked to sepsis in addition to validating the model's conclusions.

In the end, the complete model is made available as an intuitive web tool that enables medical practitioners to enter patient information and get precise, understandable forecasts instantly. By combining cutting-edge machine learning with useful clinical applications, this architecture improves patient outcomes and early sepsis detection.

4.2.2 DETAILED DESIGN



(Fig. 2: Image of Data Flow diagram of the proposed prediction sepsis model)

The suggested model's data flow diagram is shown in the image Fig. 2 above. It shows how the dataset is loaded, preprocessed using the previously discussed techniques in system architecture, and then separately trained and tested to produce the final sepsis prediction results.

5. IMPLEMENTATION AND TESTING

5.1. Data Collection and Preprocessing

This project's dataset, "Prediction of Sepsis," was obtained from Kaggle. In order to forecast the development of sepsis, it includes time series data of intensive care unit patients together with hourly recordings of vital signs, laboratory results, and demographic information. Predicting sepsis six hours before clinical onset is the aim in order to enable prompt medical intervention. Nearly 5% of all cases are classified as sepsis, and 95% of all cases are classified as non-sepsis. It has 44 columns, including the dependent column Sepsis Label.

Dataset Loading: The dataset (sepsis dataset.csv) was loaded using pandas for further processing. Initial verification was performed to ensure successful data loading.

Column Dropping: Unnamed_0, HospAdmTime, ICULOS, Hour, Unit1, and Unit2 were among the unnecessary columns that were eliminated since they did not add to the predictive task. Columns containing completely null values were eliminated to maintain dataset integrity. Columns with more than 95% missing values were discarded using a threshold mechanism (Threshold: Columns with missing values exceeding 95% of the dataset length were dropped.). The remaining columns were listed to verify data consistency.

Identification of Numeric and Categorical Columns: Features were categorized as numeric (int64, float64) or categorical (object, category). Separate imputation strategies were applied for these two feature types. Missing values in numeric columns were handled using IterativeImputer with 10 iterations and a random state of 42 for reproducibility. Categorical features were imputed using the most_frequent strategy with SimpleImputer.

5.2. Feature Engineering and Label Preparation:

Defining Target and Feature Variables: The target variable, SepsisLabel, was separated from the feature set. Feature columns were sanitized to replace non-alphanumeric characters with underscores to avoid LightGBM errors.

Class Imbalance Handling Using ADASYN : Due to class imbalance in the dataset, ADASYN (Adaptive Synthetic Sampling) was employed to oversample the minority class. ADASYN

generated synthetic samples for underrepresented classes, ensuring a balanced distribution. Post-resampling class distribution was verified to confirm balance. The dataset was divided into training and validation sets using `train_test_split`: 80% for training and 20% for validation. Stratified sampling ensured that class distribution remained consistent across splits. `Train_test_split` was used to separate the dataset into training and validation sets, with 80% going to training and 20% to validation. The class distribution was maintained across splits thanks to stratified sampling.

5.3. Model Development and Training

Base Model -An ensemble learning technique called the Random Forest Classifier (RFC) constructs several decision trees using arbitrary data subsets and aggregates the predictions to improve accuracy and lessen overfitting. Independent predictions are made by each tree, and majority vote determines the final result (for categorisation). RFC minimises variance by averaging several models and works very well with high dimensional data. It may, however, have trouble with class imbalance, which frequently results in inferior recall for the minority class (sepsis) but higher accuracy. RFC alone had somewhat lower sensitivity in predicting sepsis cases, despite its robustness, which made it less suitable for crucial applications needing high recall. A `RandomForestClassifier` with the following parameters was used:

- `n_estimators=100`: Number of trees.
- `max_depth=10`: Tree depth to prevent overfitting.
- `random_state=42`: Ensured reproducibility.

The training data (`X_train`, `y_train`) was used to train the model. Classification metrics were calculated and predictions were created using the validation data (`X_valid`).

Meta-Model - LightGBM Classifier- LightGBM (LGBM) is a gradient-boosting framework that builds trees leaf-wise, making it more efficient and faster than traditional boosting algorithms. It handles large datasets with higher accuracy and lower computational time by using histogram based algorithms to speed up training. LGBM demonstrated better handling of class imbalance, improving recall and precision. It outperformed RFC in capturing complex relationships between features and delivered better predictive performance, particularly for minority class instances. `LGBMClassifier` was selected as the meta-model with the following parameters:

- `n_estimators=100`: Number of boosting rounds.
- `max_depth=10`: Limits the tree depth.
- `reg_alpha=0.1` and `reg_lambda=0.1`: Regularization parameters to control model complexity.
- `random_state=42`: Ensured consistent results.

Stacking Classifier - Ensemble Model: In this project, Random Forest Classifier (RFC) was used as the base model and LightGBM (LGBM) as the meta-model to improve predictive performance. Because RFC can handle high dimensional data, reduce overfitting through bagging, and produce accurate predictions, it was selected as the basis model. Interpretability is further improved by its feature importance capabilities. Because of its effectiveness in identifying intricate patterns, refining predictions, and lowering errors from the basic model, LGBM was chosen as the meta-model. This combination ensures improved accuracy and resilience in sepsis prediction by utilising the superior generalisation of LGBM and the feature learning characteristics of RFC.

- **Stacking Configuration:**
 - Base model: Random Forest Classifier.
 - Meta-model: LightGBM Classifier.
 - `passthrough=False` to prevent using original features in the meta-model.

The resampled training data was used to train the stacking model, while `X_valid` was used for validation.

5.4. Model Saving and Deployment Using Flask

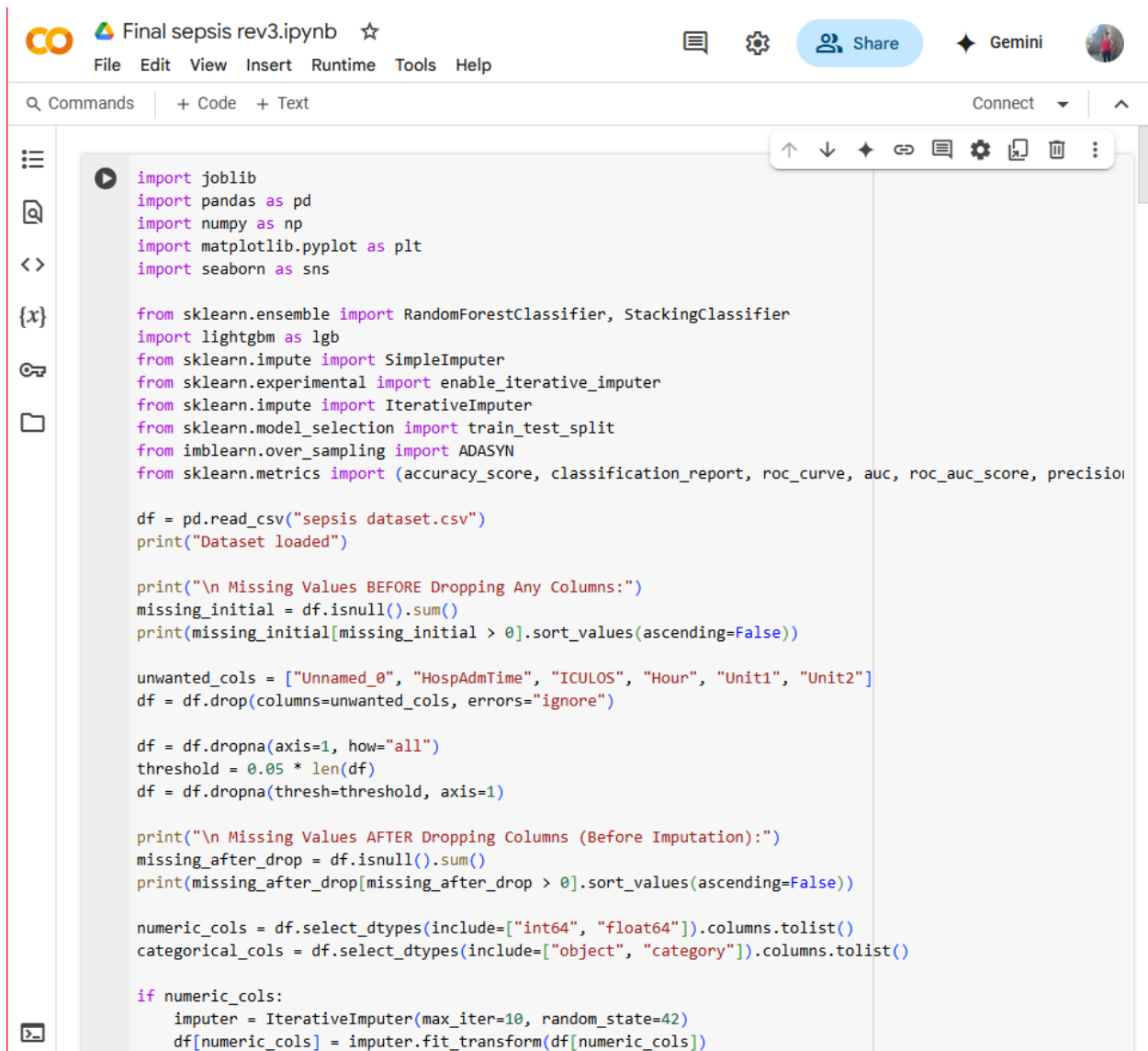
Following model evaluation and training, joblib was used to store the top-performing model for deployment. A web application built with Flask was created to let users enter medical information and get predictions in real time.

Frontend Design: A simple and intuitive interface was built using HTML, CSS, and JavaScript. Users can upload patient data or manually enter values to get the sepsis prediction.

Backend Processing: Flask API processes input data and uses the saved model to generate predictions. Results are returned to the frontend for display.

IMPLEMENTATION

BACK-END ML MODEL



```
import joblib
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.ensemble import RandomForestClassifier, StackingClassifier
import lightgbm as lgb
from sklearn.impute import SimpleImputer
from sklearn.experimental import enable_iterative_imputer
from sklearn.impute import IterativeImputer
from sklearn.model_selection import train_test_split
from imblearn.over_sampling import ADASYN
from sklearn.metrics import (accuracy_score, classification_report, roc_curve, auc, roc_auc_score, precision_score)

df = pd.read_csv("sepsis_dataset.csv")
print("Dataset loaded")

print("\n Missing Values BEFORE Dropping Any Columns:")
missing_initial = df.isnull().sum()
print(missing_initial[missing_initial > 0].sort_values(ascending=False))

unwanted_cols = ["Unnamed_0", "HospAdmTime", "ICULOS", "Hour", "Unit1", "Unit2"]
df = df.drop(columns=unwanted_cols, errors="ignore")

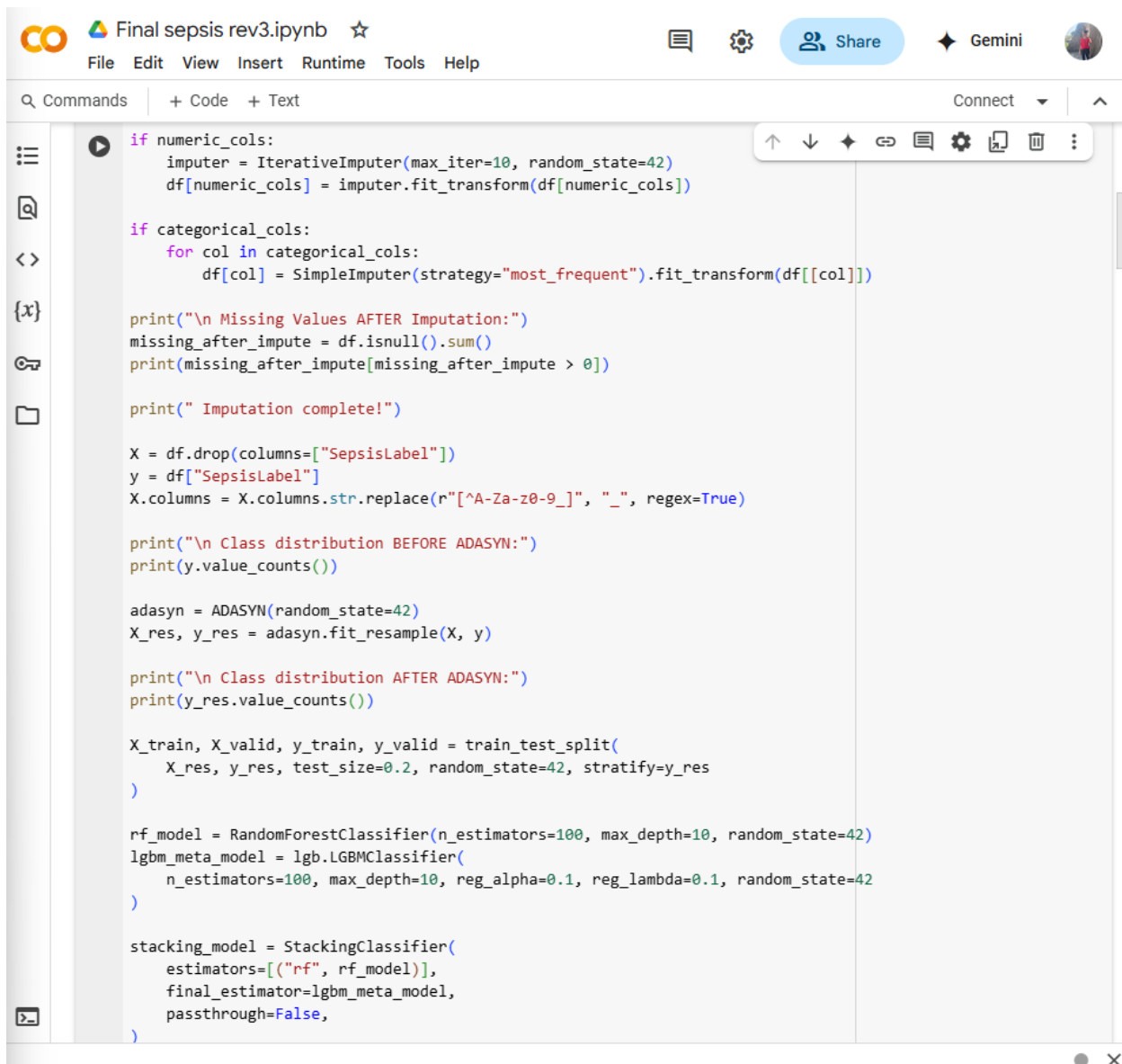
df = df.dropna(axis=1, how="all")
threshold = 0.05 * len(df)
df = df.dropna(thresh=threshold, axis=1)

print("\n Missing Values AFTER Dropping Columns (Before Imputation):")
missing_after_drop = df.isnull().sum()
print(missing_after_drop[missing_after_drop > 0].sort_values(ascending=False))

numeric_cols = df.select_dtypes(include=["int64", "float64"]).columns.tolist()
categorical_cols = df.select_dtypes(include=["object", "category"]).columns.tolist()

if numeric_cols:
    imputer = IterativeImputer(max_iter=10, random_state=42)
    df[numeric_cols] = imputer.fit_transform(df[numeric_cols])
```

(Fig. 3.1: CODE implementation- dataset loading)



```
if numeric_cols:
    imputer = IterativeImputer(max_iter=10, random_state=42)
    df[numeric_cols] = imputer.fit_transform(df[numeric_cols])

if categorical_cols:
    for col in categorical_cols:
        df[col] = SimpleImputer(strategy="most_frequent").fit_transform(df[[col]])

print("\n Missing Values AFTER Imputation:")
missing_after_impute = df.isnull().sum()
print(missing_after_impute[missing_after_impute > 0])

print(" Imputation complete!")

X = df.drop(columns=["SepsisLabel"])
y = df["SepsisLabel"]
X.columns = X.columns.str.replace(r"^[A-Za-z0-9_]", "_", regex=True)

print("\n Class distribution BEFORE ADASYN:")
print(y.value_counts())

adasyn = ADASYN(random_state=42)
X_res, y_res = adasyn.fit_resample(X, y)

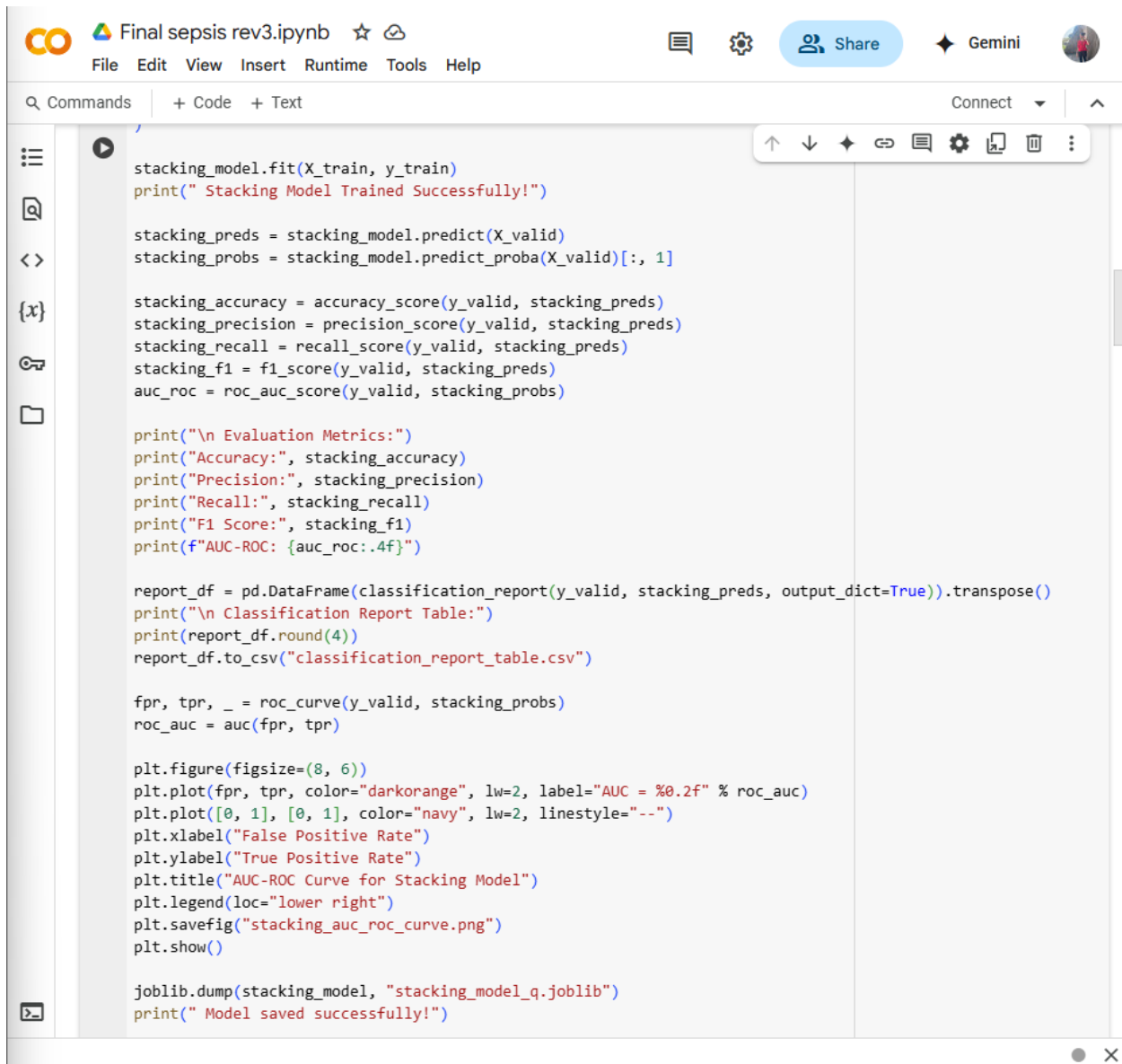
print("\n Class distribution AFTER ADASYN:")
print(y_res.value_counts())

X_train, X_valid, y_train, y_valid = train_test_split(
    X_res, y_res, test_size=0.2, random_state=42, stratify=y_res
)

rf_model = RandomForestClassifier(n_estimators=100, max_depth=10, random_state=42)
lgbm_meta_model = lgb.LGBMClassifier(
    n_estimators=100, max_depth=10, reg_alpha=0.1, reg_lambda=0.1, random_state=42
)

stacking_model = StackingClassifier(
    estimators=[("rf", rf_model)],
    final_estimator=lgbm_meta_model,
    passthrough=False,
)
```

(Fig. 3.2: CODE implementation-data preprocessing)



```
stacking_model.fit(X_train, y_train)
print(" Stacking Model Trained Successfully!")

stacking_preds = stacking_model.predict(X_valid)
stacking_probs = stacking_model.predict_proba(X_valid)[:, 1]

stacking_accuracy = accuracy_score(y_valid, stacking_preds)
stacking_precision = precision_score(y_valid, stacking_preds)
stacking_recall = recall_score(y_valid, stacking_preds)
stacking_f1 = f1_score(y_valid, stacking_preds)
auc_roc = roc_auc_score(y_valid, stacking_probs)

print("\n Evaluation Metrics:")
print("Accuracy:", stacking_accuracy)
print("Precision:", stacking_precision)
print("Recall:", stacking_recall)
print("F1 Score:", stacking_f1)
print(f"AUC-ROC: {auc_roc:.4f}")

report_df = pd.DataFrame(classification_report(y_valid, stacking_preds, output_dict=True)).transpose()
print("\n Classification Report Table:")
print(report_df.round(4))
report_df.to_csv("classification_report_table.csv")

fpr, tpr, _ = roc_curve(y_valid, stacking_probs)
roc_auc = auc(fpr, tpr)

plt.figure(figsize=(8, 6))
plt.plot(fpr, tpr, color="darkorange", lw=2, label="AUC = %0.2f" % roc_auc)
plt.plot([0, 1], [0, 1], color="navy", lw=2, linestyle="--")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("AUC-ROC Curve for Stacking Model")
plt.legend(loc="lower right")
plt.savefig("stacking_auc_roc_curve.png")
plt.show()

joblib.dump(stacking_model, "stacking_model_q.joblib")
print(" Model saved successfully!")
```

(Fig. 3.3: CODE implementation-model training and saving)

The above images were the implementation steps taken to successfully obtain this project, which includes the back end : stacking model with base model as Random forest and meta model as LightGBM and these were added to front end :Flask application that has an index page to take the input from the user in an web application and a result page to show the prediction output for the test data.

6. RESULTS AND DISCUSSION

The dataset was successfully loaded, and initial preprocessing steps were applied to ensure data quality. Columns with more than 95% missing values were dropped, leaving a set of relevant features essential for sepsis prediction. In addition to laboratory indicators like base excess, bicarbonate (HCO_3), creatinine, glucose, and platelet count, the features that were kept included vital signs like heart rate (HR), oxygen saturation (O_2Sat), temperature (Temp), systolic blood pressure (SBP), mean arterial pressure (MAP), diastolic blood pressure (DBP), and respiratory rate (Resp). The goal variable (SepsisLabel), distinct patient identities (Patient_ID), and demographic data, such as age and gender, were also maintained.

To address missing values, an Iterative Imputer was applied, leveraging a multivariate approach to predict and fill gaps in the dataset. Despite a convergence warning indicating that the early stopping criterion was not met, the imputation process was successfully completed. This ensured that all missing values were adequately handled, making the dataset suitable for subsequent model training.

With sepsis cases being under represented in the dataset, a major class imbalance was one of the main issues faced. In order to counteract this, synthetic samples for the minority class were created using ADASYN (Adaptive Synthetic Sampling). With 1,027,767 samples representing non-sepsis patients and 1,026,756 samples representing sepsis cases, the class distribution was therefore well balanced. The algorithm was able to learn from a variety of scenarios thanks to this balanced dataset, which decreased bias and enhanced its capacity to correctly identify sepsis cases.

The development of a stacking ensemble model improved overall predictive performance by utilising the advantages of several base models. One of the ensemble's fundamental models, LightGBM, performed especially well when processing massive amounts of data. LightGBM used 255 bins for feature processing during training, reporting 821,405 positive samples and 822,213 negative samples. LightGBM automatically chose column-wise multi-threading to lower computational cost and increase training efficiency when the model was initialised with a score of -0.000983.

The efficacy of the stacking model in detecting sepsis was demonstrated when its performance was assessed using common classification metrics. With an accuracy of 91.64%, the model

demonstrated a high level of overall categorisation accuracy. The model's accuracy in detecting true positives while reducing false positives was demonstrated by its precision score of 91.34%. While the F1 score of 91.67% validated a balanced trade-off between precision and recall, as shown in figure 2 (TABLE I-Evaluation metrics), the recall score of 91.99% demonstrated the model's capacity to capture the majority of real sepsis cases. Additionally, the model's outstanding discriminatory power was proved by the AUC-ROC (Area Under the Receiver Operating Characteristic Curve) value of 0.98, which is displayed in figure 1. This ensures high sensitivity and specificity in differentiating between cases of sepsis and non-sepsis.

Once training and evaluation were completed successfully, the trained stacking model was stored for further use. Other performance indicators, including the AUC-ROC curve, were also produced and downloaded for additional study and record keeping. These findings establish the suggested model as a useful tool for clinical applications by demonstrating its robustness and dependability in correctly predicting sepsis. The stacking ensemble's excellent performance indicates that it can help medical personnel make prompt, well informed judgements, which will eventually improve patient outcomes when managing sepsis.

Using ICU patient data from two hospitals, the base research work concentrated on sepsis detection models (Dataset A: 20,336 subjects, Dataset B: 20,000 subjects). It made use of logistic regression and random forest classifiers, which were assessed using a utility function that rewarded timely detections and penalised early and late predictions. While Dataset B had an accuracy of 89.23% and a utility score of 22.13%, Dataset A had an accuracy of 81.49% and a utility score of 56.85%. Despite comparatively better recall scores (0.78 and 1.00), the precision for identifying septic cases was poor (0.09 and 0.06), underscoring the difficulties in striking a balance between the two.

Data pretreatment for this project involved removing columns with more than 95% missing values while keeping vital information including blood pressure, heart rate, oxygen saturation, and demographics. With 1,027,767 non-sepsis and 1,026,756 sepsis samples, ADASYN addressed the notable class imbalance and an iterative imputer successfully handled missing values.

The performance of a stacking ensemble model that combined Random Forest and LightGBM was greatly enhanced. LightGBM used optimised feature processing and multi-threading to handle enormous amounts of data efficiently. The new model demonstrated superior discriminatory

capacity with 91.64% accuracy, 91.34% precision, 91.99% recall, 91.67% F1-score, and an AUC-ROC of 0.98.

Through sophisticated imputation and hyperparameter optimisation, the suggested model improved data handling and successfully addressed class imbalance while striking a better balance between precision and recall when compared to the base research. The excellent performance indicates that our model can help medical practitioners make prompt, well informed decisions, improving patient outcomes in the treatment of sepsis.

```
Dataset loaded
*
Missing Values BEFORE Dropping Any Columns:
Bilirubin_direct 1046802
TroponinI        1042999
Fibrinogen       1041025
Bilirubin_total  1034331
Alkalinephos     1032505
AST              1032237
EtCO2            1029988
Lactate          1016496
PTT              1007740
SaO2             1004495
Phosphate        1001098
Calcium          991554
HCO3             984535
Platelets        983255
Creatinine       981850
Chloride         981288
WBC              975617
Magnesium        974683
PaCO2            973484
BUN              969878
BaseExcess       965620
Hgb              963654
pH               952148
Potassium        942714
Hct              940505
FiO2             930506
Glucose          894981
Temp             694133
Unit1            461433
Unit2            461433
DBP              416366
SBP              156205
Resp            131685
O2Sat           131684
MAP             119077
HR              92222
HospAdmTime      8
dtype: int64

Missing Values AFTER Dropping Columns (Before Imputation):
Calcium          991554
HCO3             984535
Platelets        983255
Creatinine       981850
Chloride         981288
WBC              975617
Magnesium        974683
PaCO2            973484
BUN              969878
BaseExcess       965620
Hgb              963654
pH               952148
Potassium        942714
Hct              940505
FiO2             930506
Glucose          894981
Temp             694133
DBP              416366
SBP              156205
Resp            131685
O2Sat           131684
MAP             119077
HR              92222
dtype: int64
/usr/local/lib/python3.11/dist-packages/sklearn/impute/_iterat
warnings.warn(

Missing Values AFTER Imputation:
Series([], dtype: int64)
Imputation complete!
```

(Figure 4: Before and after imputation of missing values)

```

Class distribution BEFORE ADASYN:
SepsisLabel
0.0    1027767
1.0     20808
Name: count, dtype: int64

```

```

Class distribution AFTER ADASYN:
SepsisLabel
0.0    1027767
1.0    1026756
Name: count, dtype: int64

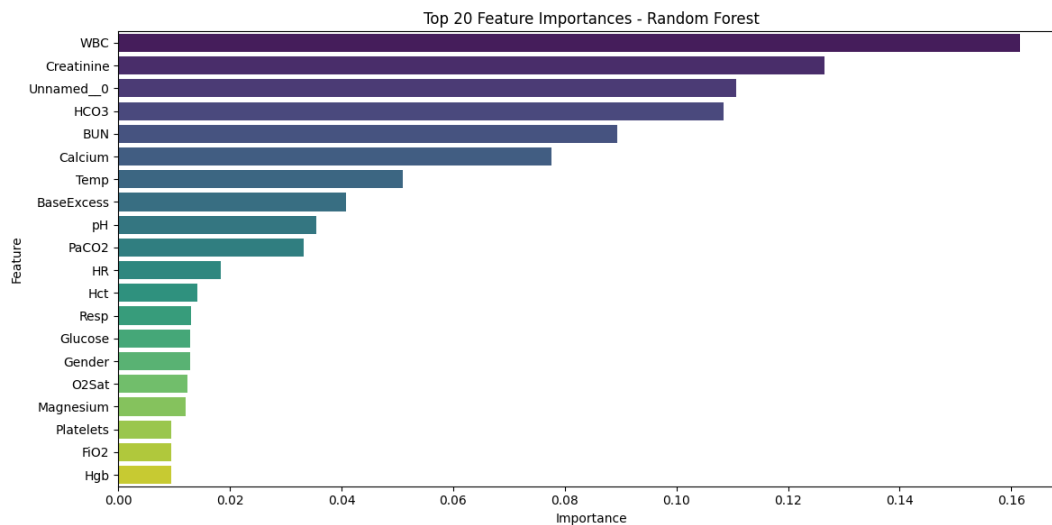
```

(Fig. 5: Result of handling class imbalance)

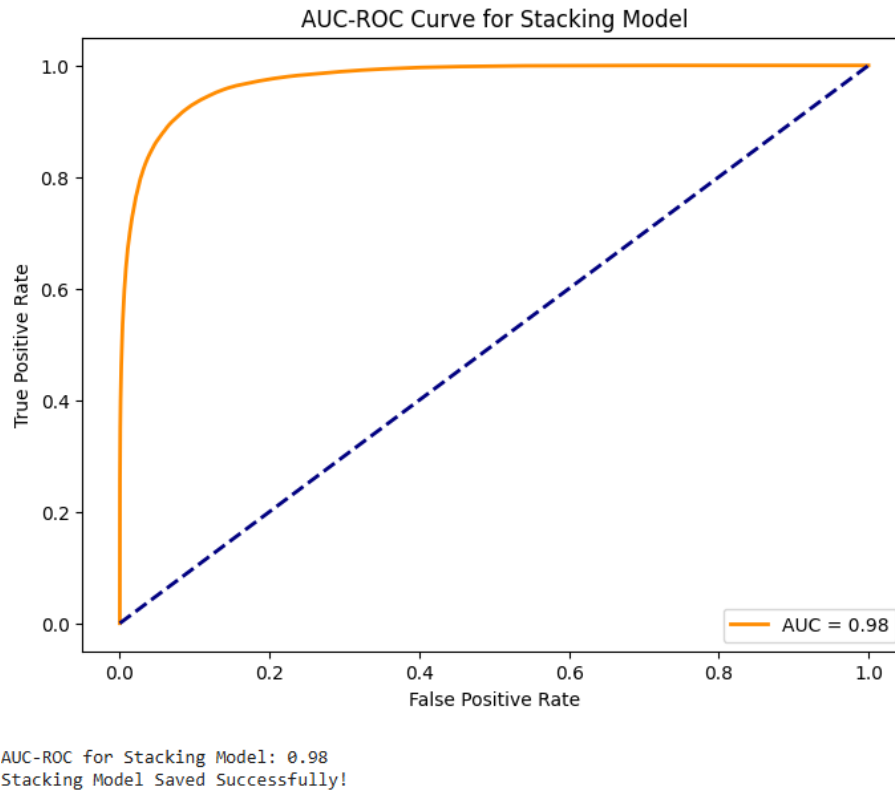
Classification Report Table:

	precision	recall	f1-score	support
0.0	0.9195	0.9129	0.9162	205554.0000
1.0	0.9134	0.9200	0.9167	205351.0000
accuracy	0.9164	0.9164	0.9164	0.9164
macro avg	0.9165	0.9164	0.9164	410905.0000
weighted avg	0.9165	0.9164	0.9164	410905.0000

(Fig. 6: Result of the evaluation metrics)

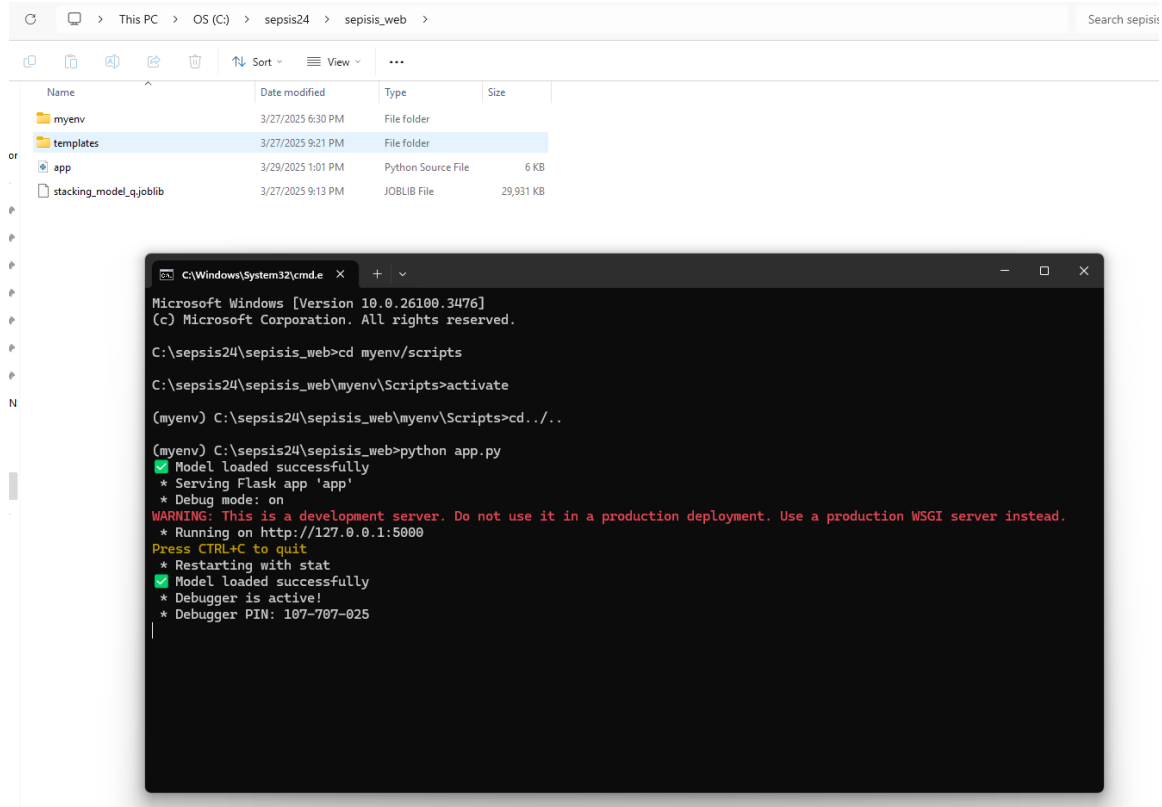


(Fig. 7: Feature importance in a bar chart)



(Fig. 8: AUC-ROC Curve)

The above model is loaded into another python file “app.py” which is a Flask application that combines the back end model and front end model. To activate this environment the below code is implemented in the command prompt. Later the local address is copied and opened in the browser to use the web application and predict on a user given data and classify it.



(Fig. 9: Implementation of the web page)

The screenshot shows a web browser window with the address bar set to '127.0.0.1:5000'. The page title is 'Sepsis Risk Assessment Tool'. The interface is divided into three main sections: Vital Signs, Blood Gas Analysis, and Laboratory Values. Each section contains a table of input fields with test values entered.

Vital Signs			
Heart Rate (bpm) *	Oxygen Saturation (%) *	Temperature (°C) *	Systolic BP (mmHg) *
95	92	38.5	110
Mean Arterial Pressure *	Diastolic BP (mmHg) *	Respiratory Rate *	End Tidal CO2 (mmHg)
75	70	22	

Blood Gas Analysis			
Base Excess (mEq/L)	Bicarbonate (mEq/L) *	FIO2	pH *
	22		7.35
PaCO2 (mmHg) *	SaO2 (%) *		
45	94		

Laboratory Values			
AST (U/L)	BUN (mg/dL) *	Alkaline Phosphatase (U/L)	Calcium (mg/dL) *
	15		9
Chloride (mEq/L) *	Creatinine (mg/dL) *	Direct Bilirubin (mg/dL)	Glucose (mg/dL) *

(Fig. 10: Test values in webpage)

The screenshot shows a web browser window with the title "Sepsis Risk Assessment". The URL bar shows "127.0.0.1:5000". The page contains several input fields for lab tests and patient information. The lab tests section includes fields for Chloride (mEq/L), Creatinine (mg/dL), Direct Bilirubin (mg/dL), Glucose (mg/dL), Lactate (mmol/L), Magnesium (mg/dL), Phosphate (mg/dL), Potassium (mEq/L), Total Bilirubin (mg/dL), and Troponin I (ng/mL). The Hematology section includes fields for Hematocrit (%), Hemoglobin (g/dL), PTT (sec), and WBC (cells/ul). The Patient section includes fields for Age (years), Gender (Male), and HR/Age Ratio. A blue button labeled "Assess Sepsis Risk" is at the bottom. A note at the bottom states: "Fields marked with * are required. Leave others blank if data is unavailable."

Lab Test	Value
Chloride (mEq/L) *	100
Creatinine (mg/dL) *	1
Direct Bilirubin (mg/dL)	
Glucose (mg/dL) *	120
Lactate (mmol/L) *	1.5
Magnesium (mg/dL) *	1.9
Phosphate (mg/dL) *	3.8
Potassium (mEq/L) *	4.2
Total Bilirubin (mg/dL) *	1.2
Troponin I (ng/mL)	

Hematology	Value
Hematocrit (%) *	38
Hemoglobin (g/dL) *	13
PTT (sec)	
WBC (cells/ul) *	13000

Patient	Value
Age (years)	40
Gender	Male
HR/Age Ratio	2.4

Assess Sepsis Risk

Fields marked with * are required. Leave others blank if data is unavailable.

(Fig. 11: Submitting test values in webpage)

The screenshot shows the "Sepsis Risk Assessment Results" page. It features a header with four risk levels: No Sepsis, SIRS/Sepsis, Severe Sepsis, and Critical. The "SIRS/Sepsis" result is highlighted in orange. Below the result, the "92.0% Model Accuracy" is displayed. A "Clinical Recommendation" section provides a warning and a list of actions: Close monitoring (q1-2h vitals), Consider sepsis workup, Repeat assessment in 2-4 hours, Consider early antibiotic therapy if infection suspected, and Monitor for signs of organ dysfunction. A blue button labeled "Assess Another Patient" is at the bottom.

Sepsis Risk Assessment Results

No Sepsis | **SIRS/Sepsis** | Severe Sepsis | Critical

SIRS/Sepsis

92.0%
Model Accuracy

Clinical Recommendation:

Warning: This patient meets SIRS criteria but no organ dysfunction detected.

- Close monitoring (q1-2h vitals)
- Consider sepsis workup
- Repeat assessment in 2-4 hours
- Consider early antibiotic therapy if infection suspected
- Monitor for signs of organ dysfunction

Assess Another Patient

(Fig. 12: Final prediction results)

7. CONCLUSION AND SCOPE FOR FUTURE WORK

This proposed work has provided a comprehensive overview of the advancements and challenges in early sepsis detection using machine learning techniques. Sepsis remains a critical global health concern, with high mortality rates necessitating timely and accurate diagnosis.

By combining Random Forest and LightGBM with sophisticated imputation and resampling approaches, the suggested ensemble based strategy outperformed conventional methods by achieving an accuracy of 91.64%, a recall of 91.99%, and an AUC-ROC of 0.98. These findings demonstrate how stacking ensemble models can improve predictive accuracy while preserving resilience in the face of data heterogeneity. Furthermore, iterative imputation for addressing missing data and ADASYN for mitigating class imbalance were essential in enhancing model reliability.

Future research is still needed to address issues like real time implementation, model interpretability, and generalisability across many clinical situations, despite recent developments. Sepsis prediction systems may be further improved by incorporating explainable AI (XAI) approaches and investigating deep learning models. In the end, this project highlights how machine learning can revolutionise the treatment of sepsis, opening the door to quicker interventions and better patient outcomes in critical care settings.

Future work should focus on deploying these models in real time clinical workflows, validating their performance across multi center datasets, applying more advanced imputation techniques ,hyper parameterization and enhancing interpretability to foster trust among healthcare professionals. By addressing these challenges, machine learning can play a pivotal role in reducing sepsis related mortality and improving global healthcare outcomes.

ANNEXURE -I – PSEUDO CODE

1. BEGIN
2. IMPORT necessary libraries (joblib, pandas, numpy, matplotlib, seaborn, sklearn, lightgbm, imblearn)
3. LOAD dataset from "sepsis dataset.csv" into dataframe df
4. PRINT missing values before any preprocessing
5. DROP unwanted columns: ["Unnamed_0", "HospAdmTime", "ICULOS", "Hour", "Unit1", "Unit2"]
6. DROP columns with all missing values
7. DROP columns with more than 5% missing data
8. PRINT missing values after dropping columns
9. IDENTIFY numeric and categorical columns from df
10. IF numeric columns exist THEN
11. APPLY IterativeImputer to fill missing numeric values
12. IF categorical columns exist THEN
13. FOR each categorical column
14. APPLY SimpleImputer with "most_frequent" strategy
15. PRINT missing values after imputation
16. SPLIT df into features X and target y (SepsisLabel)
17. CLEAN column names in X to remove special characters

18. PRINT class distribution in target y (before ADASYN)
19. APPLY ADASYN to balance classes in X and y
20. PRINT class distribution after ADASYN
21. SPLIT X and y into training and validation sets (80/20 stratified)
22. INITIALIZE:
 - RandomForestClassifier as base model
 - LGBMClassifier as meta model
23. CREATE StackingClassifier with RF as base and LGBM as final estimator
24. TRAIN stacking model on training data
25. PREDICT labels and probabilities on validation set
26. CALCULATE evaluation metrics:
 - Accuracy, Precision, Recall, F1 Score, AUC-ROC
27. PRINT all evaluation metrics
28. GENERATE classification report
29. SAVE classification report as CSV
30. COMPUTE ROC curve
31. PLOT ROC curve and SAVE as image
32. EXTRACT feature importances from trained RandomForest inside stacking model
33. SORT features by importance
34. DISPLAY top 10 important features
35. SAVE feature importances to CSV

36. PLOT top 20 feature importances using bar chart
37. SAVE feature importance plot as image
38. SAVE trained stacking model using joblib
39. TRY
40. DOWNLOAD model and output files (if in Colab)
41. EXCEPT
42. PRINT "Files saved locally"
43. END

REFERENCES

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